

Announcements

- HW3 is due **today, February 20**, 11:59pm PT
- Project 3 is due **Tuesday, February 27**, 11:59pm PT
- HW4 out later this week; due **Friday, March 1, 11:59pm PT**
- Midterm: **Tuesday, March 5, 7pm PT** (more info on website)

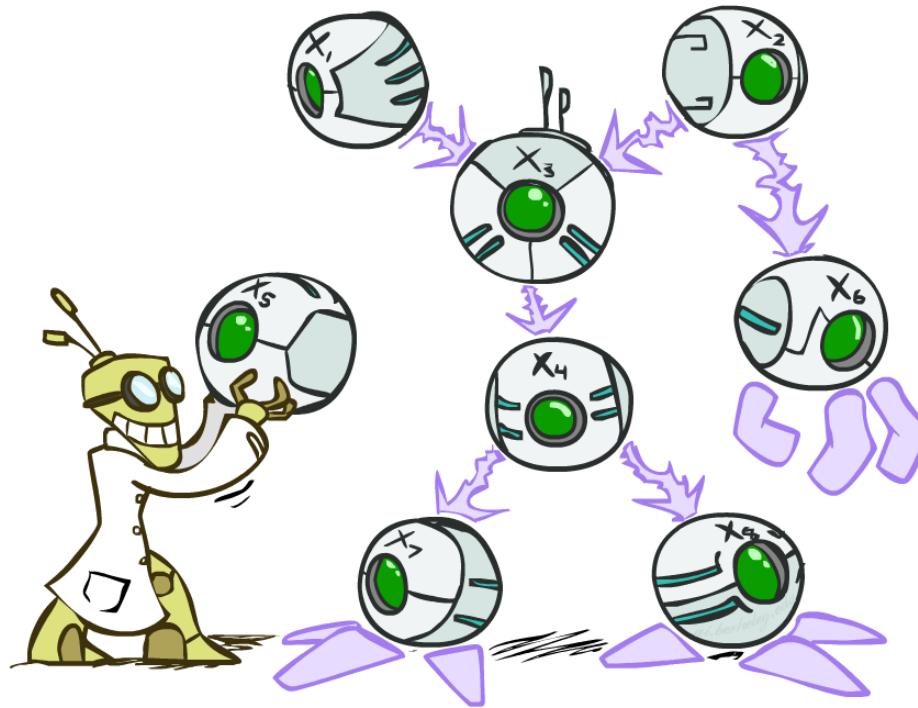


Pre-scan attendance
QR code now!

(Password appears later)

CS 188: Artificial Intelligence

Bayes Nets



Recap: Probability

- Basic laws: $0 \leq P(\omega) \leq 1$ $\sum_{\omega \in \Omega} P(\omega) = 1$
- Events: subsets of Ω : $P(A) = \sum_{\omega \in A} P(\omega)$
- Random variable $X(\omega)$ has a value in each ω
 - Distribution $P(X)$ gives probability for each possible value x
 - Joint distribution $P(X, Y)$ gives total probability for each combination x, y
- Summing out/marginalization: $P(X=x) = \sum_y P(X=x, Y=y)$
- Conditional probability: $P(X|Y) = P(X, Y)/P(Y)$
- Product rule: $P(X|Y)P(Y) = P(X, Y) = P(Y|X)P(X)$
 - Generalize to chain rule: $P(X_1, \dots, X_n) = \prod_i P(X_i | X_1, \dots, X_{i-1})$

Recap: Strict Independence

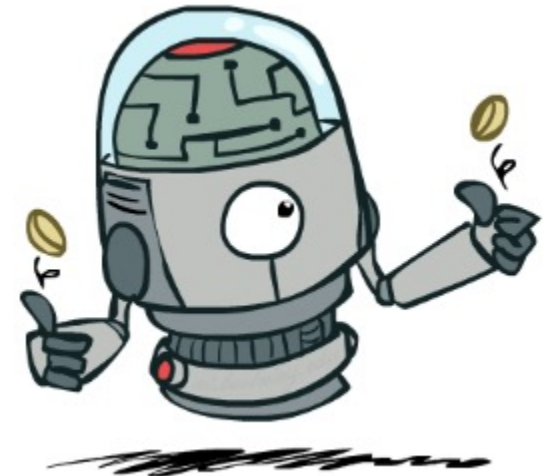
- Two variables X and Y are (absolutely) **independent** if

$$\forall x, y \quad P(x, y) = P(x) P(y)$$

- I.e., the joint distribution **factors** into a product of two simpler distributions
- Equivalently, via the product rule $P(x, y) = P(x | y) P(y)$,

$$P(x | y) = P(x) \quad \text{or} \quad P(y | x) = P(y)$$

- Example: two dice rolls $Roll_1$ and $Roll_2$
 - $P(Roll_1=5, Roll_2=3) = P(Roll_1=5) P(Roll_2=3) = 1/6 \times 1/6 = 1/36$
 - $P(Roll_2=3 | Roll_1=5) = P(Roll_2=3)$



Recap: Strict Independence

- n fair, independent coin flips:

$P(X_1)$

H	0.5
T	0.5

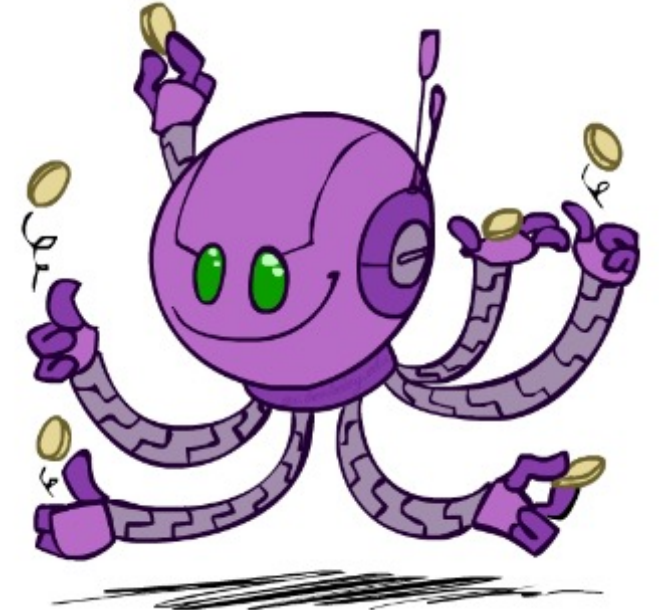
$P(X_2)$

H	0.5
T	0.5

...

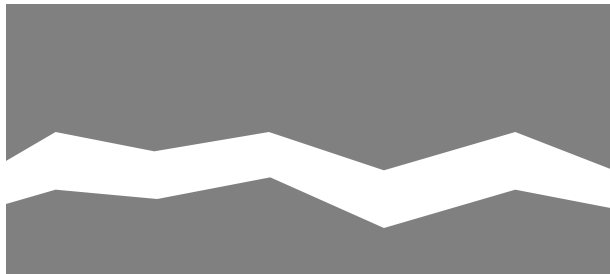
$P(X_n)$

H	0.5
T	0.5



$P(X_1, X_2, \dots, X_n)$

2^n



Example: Strict Independence



$P(\text{Rain, Traffic, Umbrella})$

Rain	Traffic	Umbrella	P	P_indep.
F	F	F	0.504	0.314
F	F	T	0.056	0.141
F	T	F	0.126	0.169
F	T	T	0.014	0.076
T	F	F	0.018	0.135
T	F	T	0.072	0.060
T	T	F	0.042	0.072
T	T	T	0.168	0.033



We compute it as is completely different

$P(\text{Rain})$ $P(\text{Traffic})$ $P(\text{Umbrella})$

F	T		F	T		F	T	
0.7	0.3	*	0.65	0.35	*	0.69	0.31	=

Example: Chain Rule



$P(\text{Rain, Traffic, Umbrella})$

Rain	Traffic	Umbrella	P
F	F	F	0.504
F	F	T	0.056
F	T	F	0.126
F	T	T	0.014
T	F	F	0.018
T	F	T	0.072
T	T	F	0.042
T	T	T	0.168



$P(\text{Rain})$

F	T
0.7	0.3

*

$P(\text{Traf.} | \text{Rain})$

	Traffic	
Rain	F	T
F	0.8	0.2
T	0.3	0.7

*

$P(\text{Umbr.} | \text{Rain, Traf.})$

		Umbrella	
Rain	Traffic	F	T
F	F	0.9	0.1
F	T	0.9	0.1
T	F	0.2	0.8
T	T	0.2	0.8

conditional independence

=

We should use this

Example: Chain Rule



$P(\text{Rain, Traffic, Umbrella})$

Rain	Traffic	Umbrella	P
F	F	F	0.504
F	F	T	0.056
F	T	F	0.126
F	T	T	0.014
T	F	F	0.018
T	F	T	0.072
T	T	F	0.042
T	T	T	0.168



$P(\text{Traf.} | \text{Rain})$

$P(\text{Umbr.} | \text{Rain})$

$P(\text{Rain})$

F	T
0.7	0.3

*

	Traffic	
Rain	F	T
F	0.8	0.2
T	0.3	0.7

*

	Umbrella	
Rain	F	T
F	0.9	0.1
T	0.2	0.8

conditional independence

=

Conditional Independence

- ***Conditional independence*** is our most basic and robust form of knowledge about uncertain environments.

- X is conditionally independent of Y given Z if and only if:

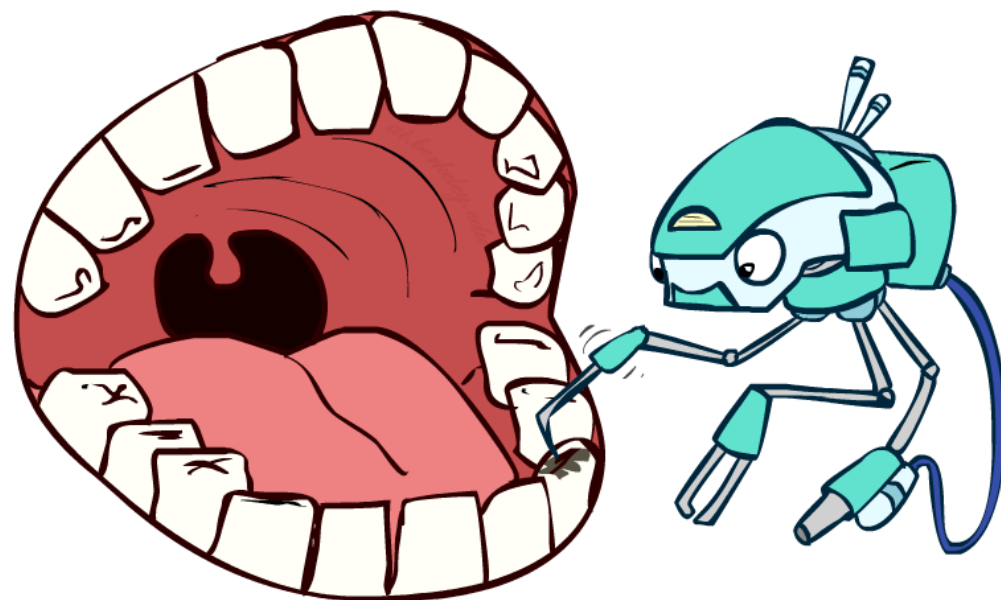
$$\forall x, y, z \quad P(x \mid y, z) = P(x \mid z)$$

or, equivalently, if and only if

$$\forall x, y, z \quad P(x, y \mid z) = P(x \mid z) P(y \mid z)$$

Conditional Independence

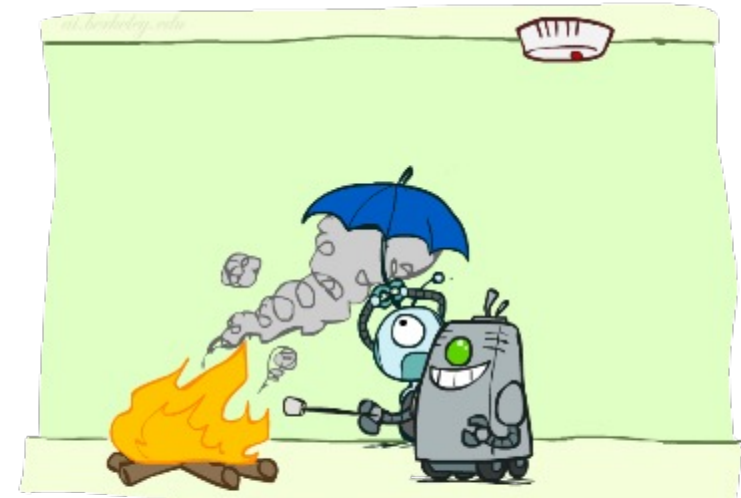
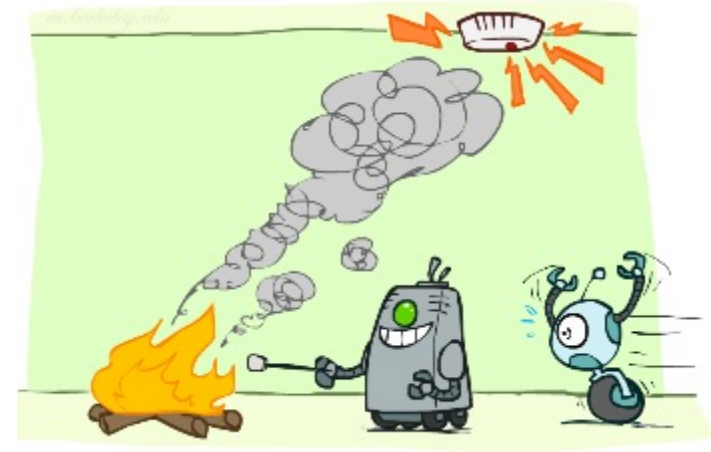
- $P(\text{Toothache}, \text{Cavity}, \text{Catch})$
- If I have a cavity, the probability that the probe catches in it doesn't depend on whether I have a toothache:
 - $P(+\text{catch} \mid +\text{toothache}, +\text{cavity}) = P(+\text{catch} \mid +\text{cavity})$
- The same independence holds if I don't have a cavity:
 - $P(+\text{catch} \mid +\text{toothache}, -\text{cavity}) = P(+\text{catch} \mid -\text{cavity})$
- Catch is *conditionally independent* of Toothache given Cavity:
 - $P(\text{Catch} \mid \text{Toothache}, \text{Cavity}) = P(\text{Catch} \mid \text{Cavity})$
- Equivalent statements:
 - $P(\text{Toothache} \mid \text{Catch}, \text{Cavity}) = P(\text{Toothache} \mid \text{Cavity})$
 - $P(\text{Toothache}, \text{Catch} \mid \text{Cavity}) = P(\text{Toothache} \mid \text{Cavity}) P(\text{Catch} \mid \text{Cavity})$
 - One can be derived from the other easily



Conditional Independence

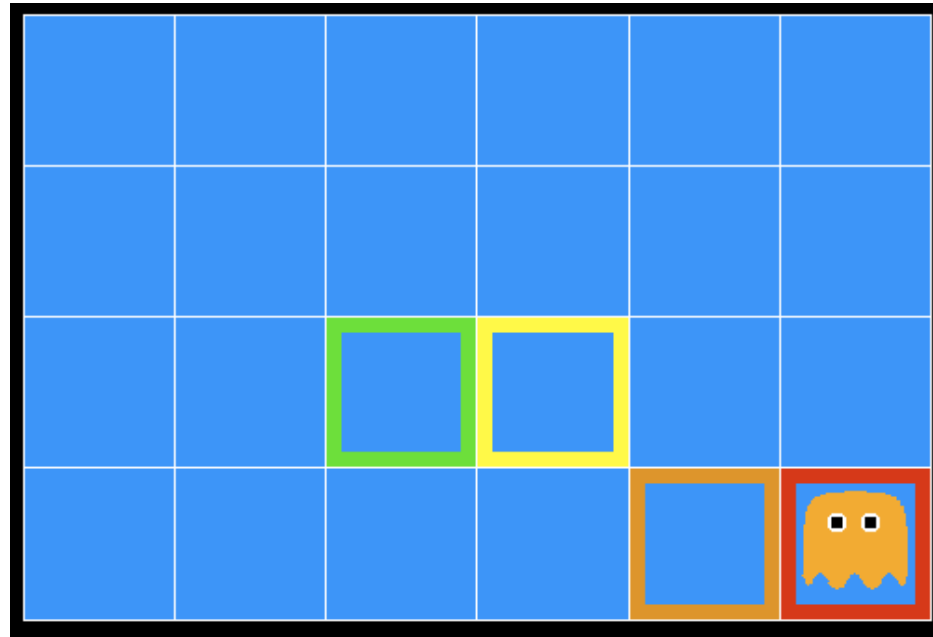
- What about this domain:

- Fire
- Smoke
- Alarm



Ghostbusters

- A ghost is in the grid somewhere
- Sensor readings tell how close a square is to the ghost
 - On the ghost: usually red
 - 1 or 2 away: usually orange
 - 3 or 4 away: usually yellow
 - 5+ away: usually green
- Click on squares until confident of location, then “*bust*”



Video of Demo Ghostbusters with Probability



Ghostbusters model

- Variables and ranges:

- G (ghost location) in $\{(1,1), \dots, (3,3)\}$
- $C_{x,y}$ (color measured at square x,y) in $\{\text{red}, \text{orange}, \text{yellow}, \text{green}\}$

0.11	0.11	0.11
0.11	0.11	0.11
0.11	0.11	0.11

- Ghostbuster physics:

- **Uniform prior distribution** over ghost location: $P(G)$
- **Sensor model**: $P(C_{x,y} \mid G)$ (depends only on distance to G)
 - E.g. $P(C_{1,1} = \text{yellow} \mid G = (1,1)) = 0.1$

Ghostbusters model, contd.

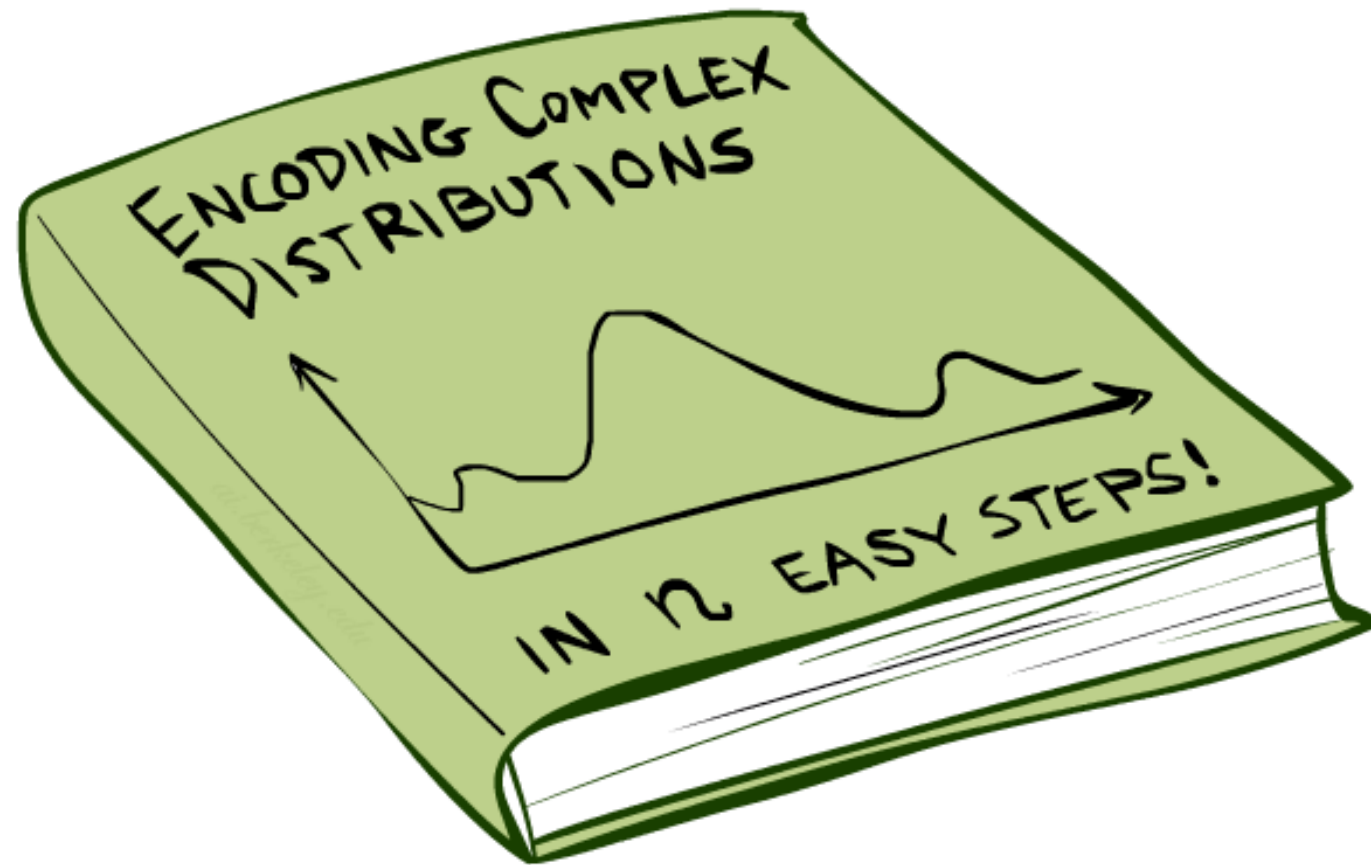
- $P(G, C_{1,1}, \dots, C_{3,3})$ has $9 \times 4^9 = 2,359,296$ entries!!!
- Ghostbuster independence:
 - Are $C_{1,1}$ and $C_{1,2}$ independent?
 - E.g., does $P(C_{1,1} = \text{yellow}) = P(C_{1,1} = \text{yellow} \mid C_{1,2} = \text{orange})$?
- Ghostbuster physics again:
 - $P(C_{x,y} \mid G)$ ***depends only on distance to G***
 - So $P(C_{1,1} = \text{yellow} \mid \underline{G = (2,3)}) = P(C_{1,1} = \text{yellow} \mid \underline{G = (2,3)}, C_{1,2} = \text{orange})$
 - I.e., $C_{1,1}$ is ***conditionally independent*** of $C_{1,2}$ ***given G***

0.11		0.11
0.11	0.11	0.11
0.11	0.11	0.11

Ghostbusters model, contd.

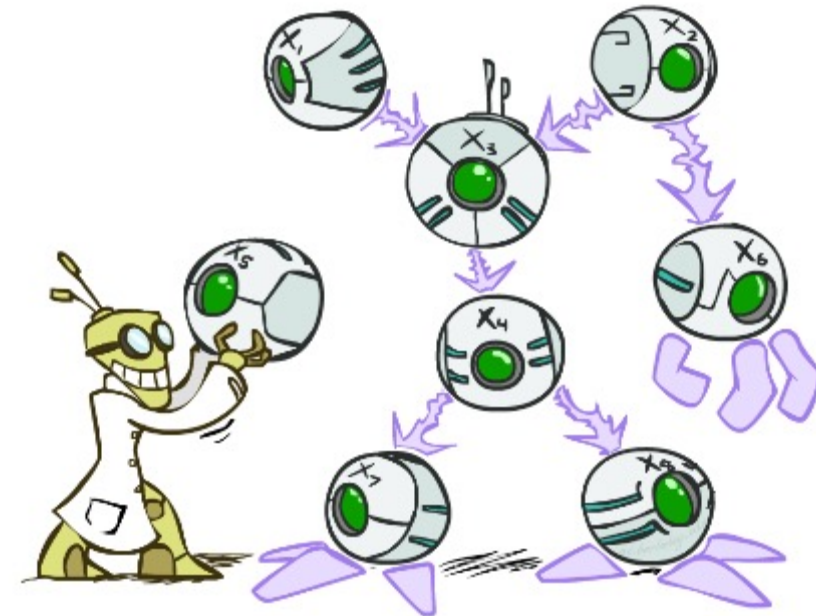
- Apply the chain rule to decompose the joint probability model:
- $P(G, C_{1,1}, \dots, C_{3,3}) = P(G) P(C_{1,1} \mid G) P(C_{1,2} \mid G, C_{1,1}) P(C_{1,3} \mid G, C_{1,1}, C_{1,2}) \dots P(C_{3,3} \mid G, C_{1,1}, \dots, C_{3,2})$
- Now simplify using conditional independence:
- $P(G, C_{1,1}, \dots, C_{3,3}) = P(G) P(C_{1,1} \mid G) P(C_{1,2} \mid G) P(C_{1,3} \mid G) \dots P(C_{3,3} \mid G)$
- I.e., conditional independence properties of ghostbuster physics simplify the probability model from *exponential* to *quadratic* in the number of squares

Bayes Nets: Big Picture



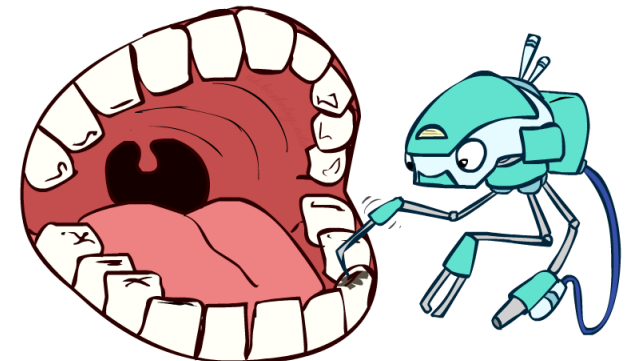
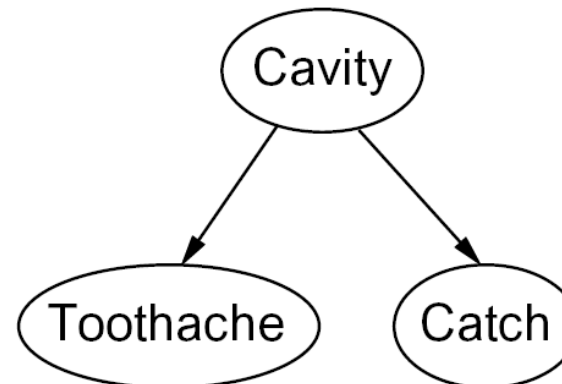
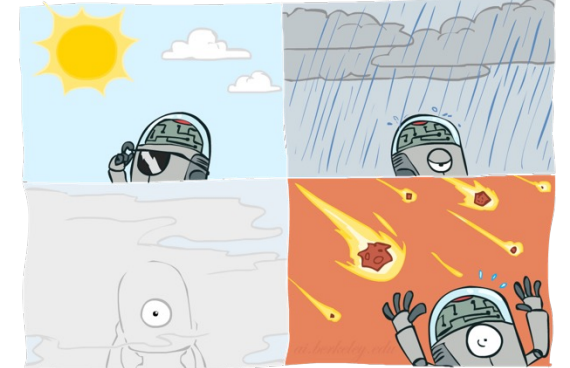
Bayes Nets: Big Picture

- **Bayes nets:** a technique for describing complex joint distributions (models) using simple, conditional distributions
 - A subset of the general class of **graphical models**
- Use local causality/conditional independence:
 - the world is composed of many variables,
 - each interacting locally with a few others
- Outline
 - Representation
 - Exact inference
 - Approximate inference



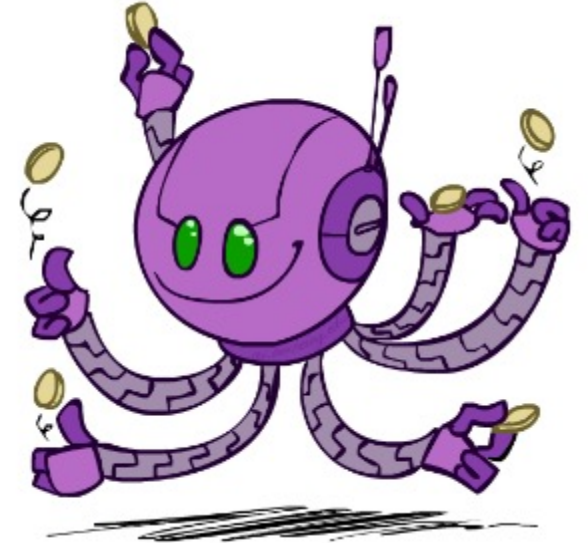
Graphical Model Notation

- **Nodes: variables (with domains)**
 - Can be assigned (observed) or unassigned (unobserved)
- **Arcs: interactions**
 - Indicate “direct influence” between variables
 - Formally: absence of arc encodes conditional independence (more later)
- For now: imagine that arrows mean direct causation (in general, they don't!)



Example: Coin Flips

- n independent coin flips

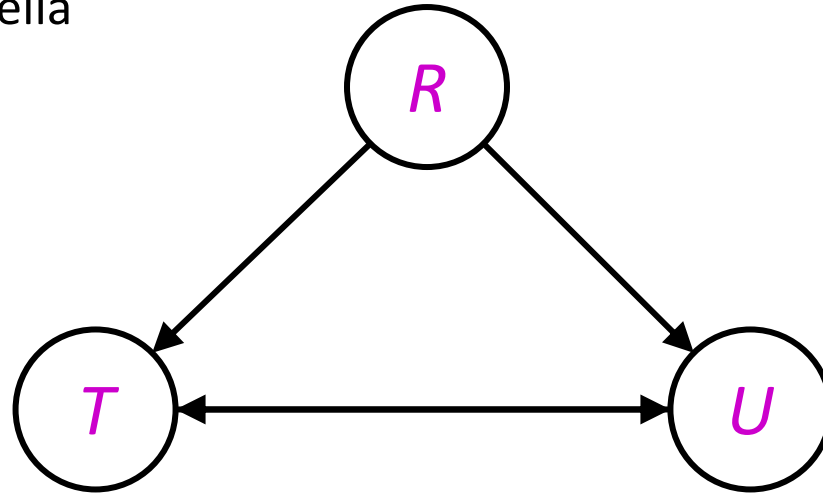


- No interactions between variables: strict independence

Example: Traffic

- Variables:

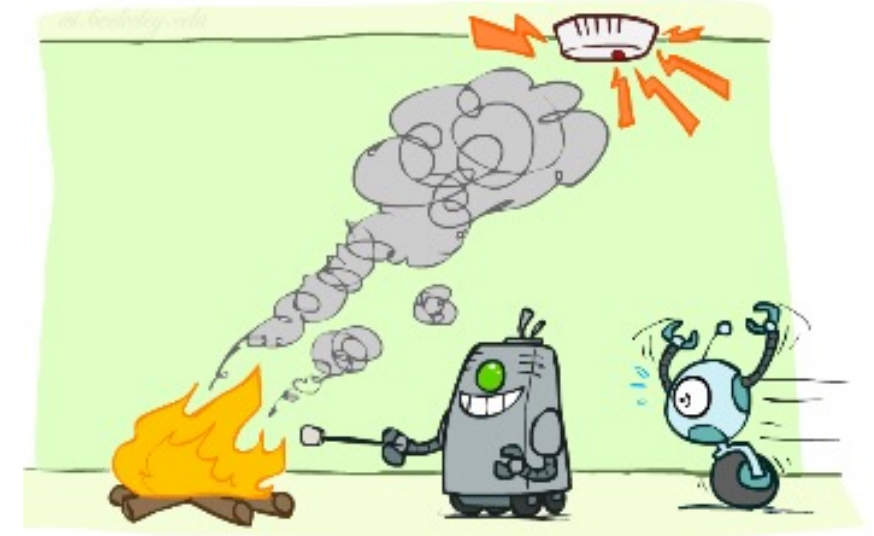
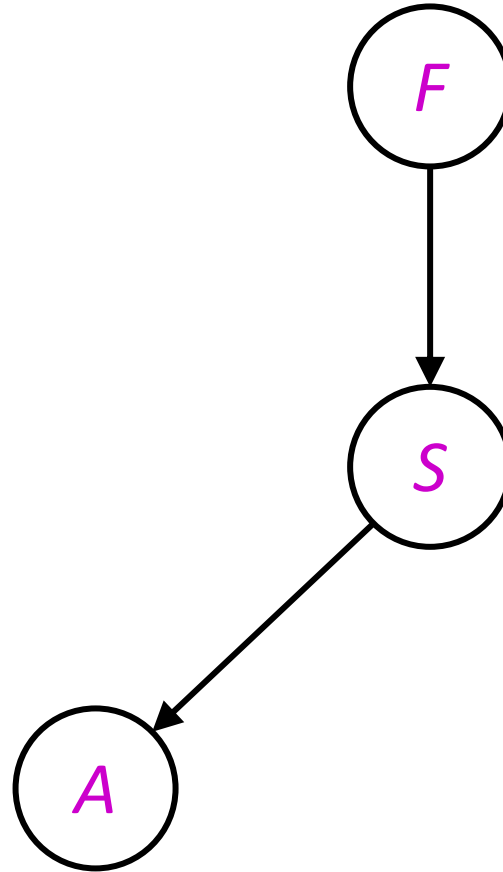
- T : There is traffic
- U : I'm holding my umbrella
- R : It rains



Example: Smoke alarm

- Variables:

- **F**: There is fire
- **S**: There is smoke
- **A**: Alarm sounds



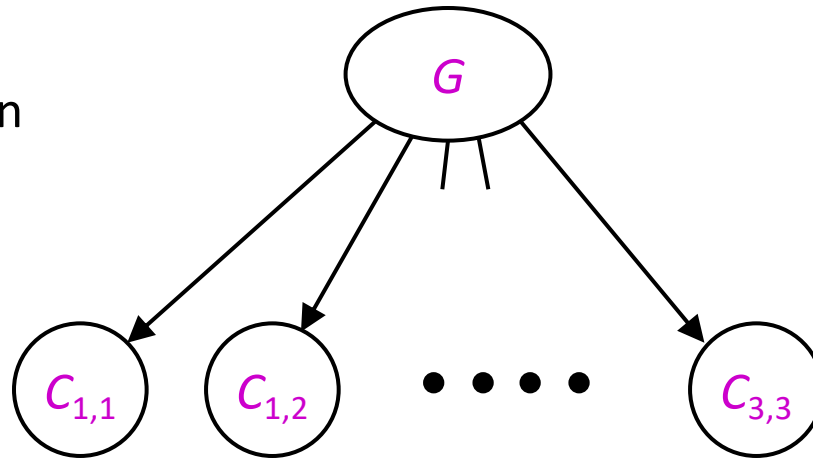
Example: Ghostbusters

- Variables:

- G : The ghost's location
- $C_{1,1}, \dots, C_{3,3}$:
The observation at each location

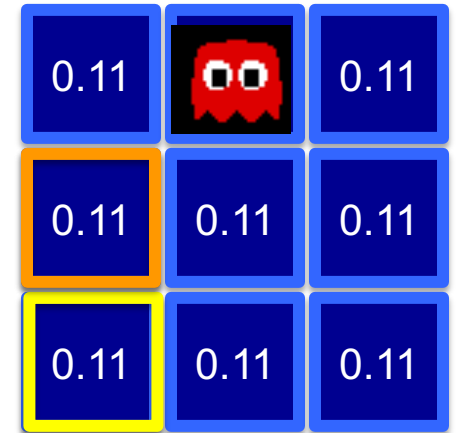
- Want to estimate:

$$P(G \mid C_{1,1}, \dots, C_{3,3})$$

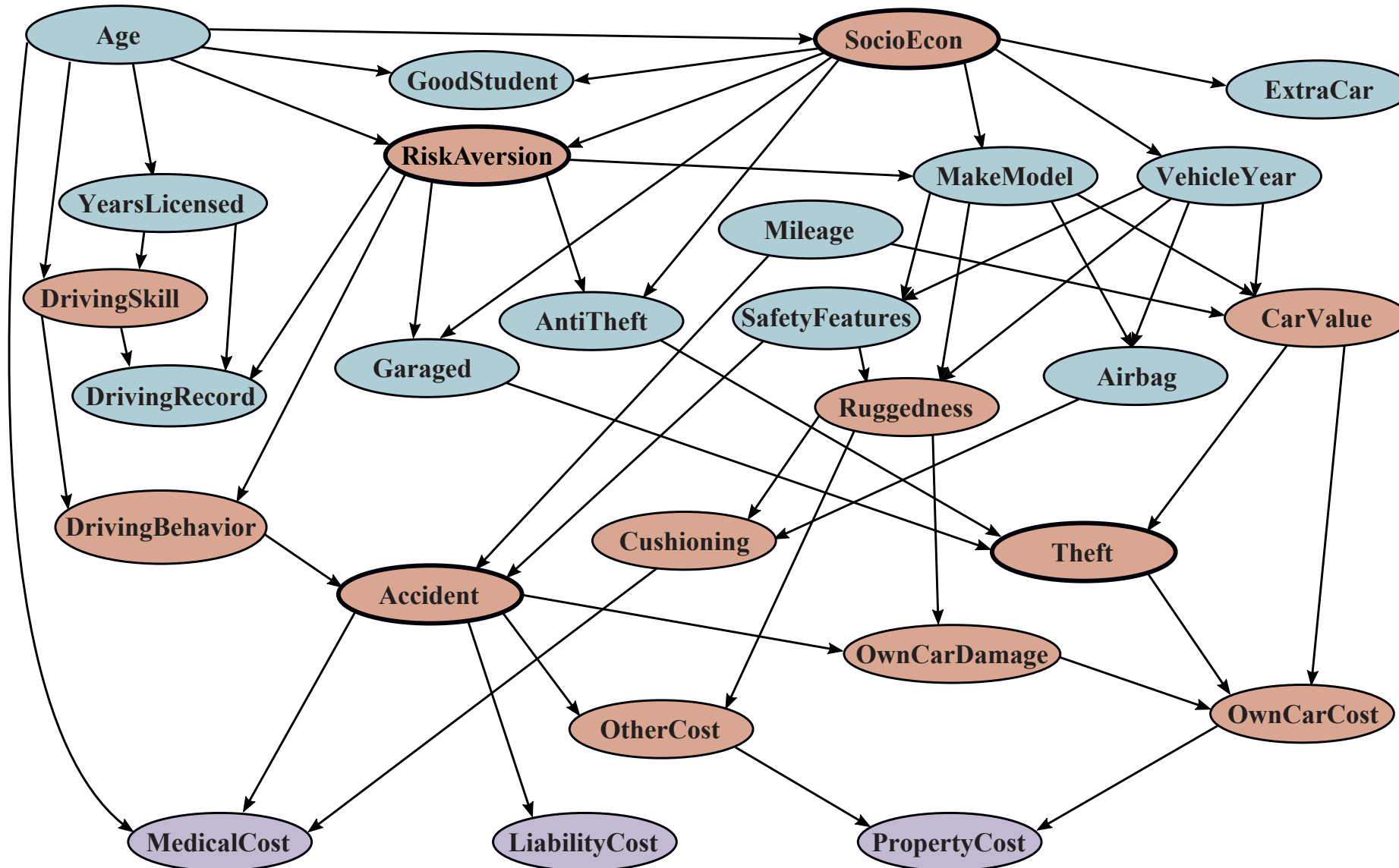


- This is called a **Naïve Bayes** model:

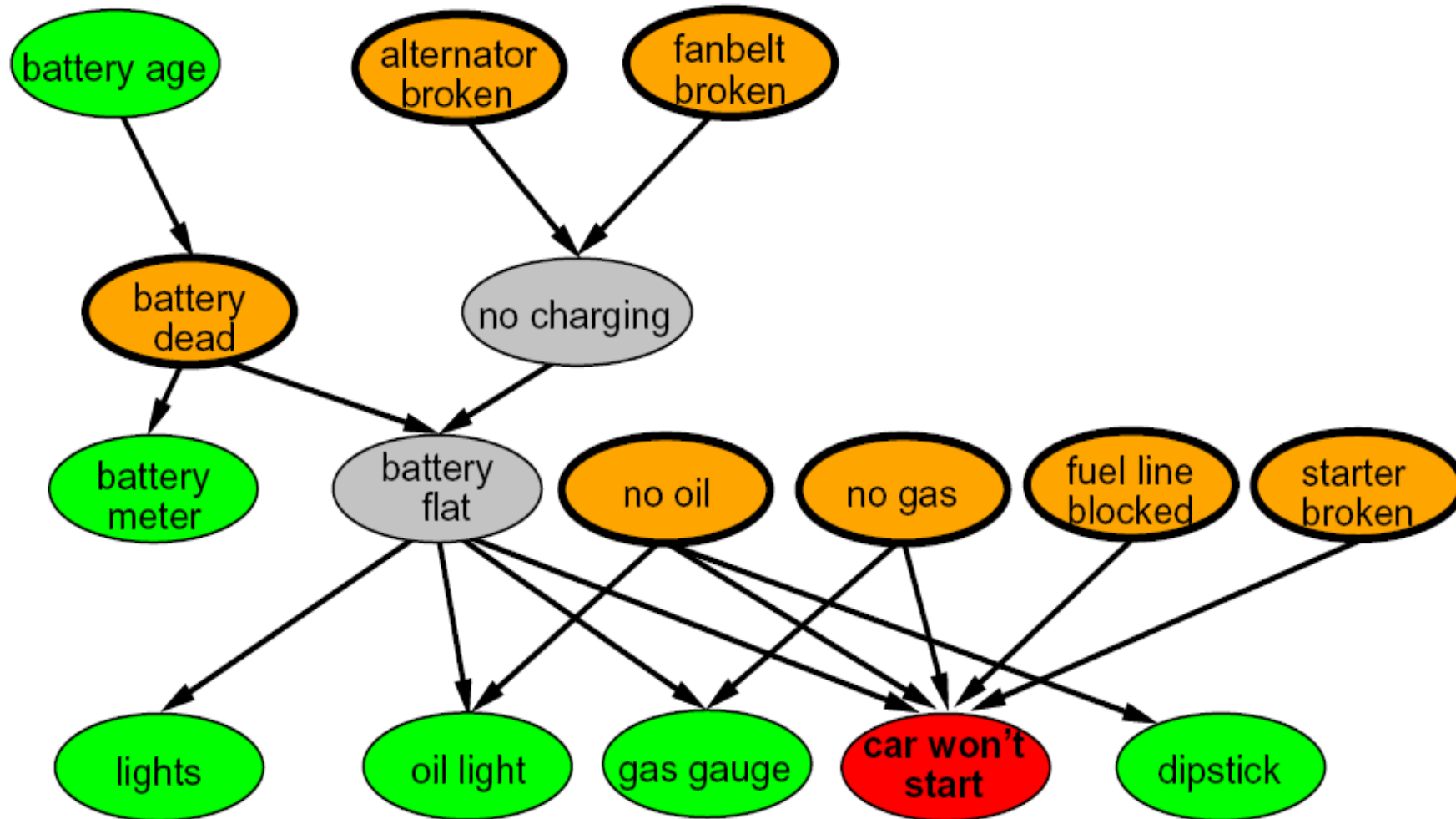
- One discrete query variable (often called the **class** or **category** variable)
- All other variables are (potentially) evidence variables
- Evidence variables are all conditionally independent given the query variable



Example Bayes' Net: Car Insurance



Example Bayes' Net: Car Won't Start



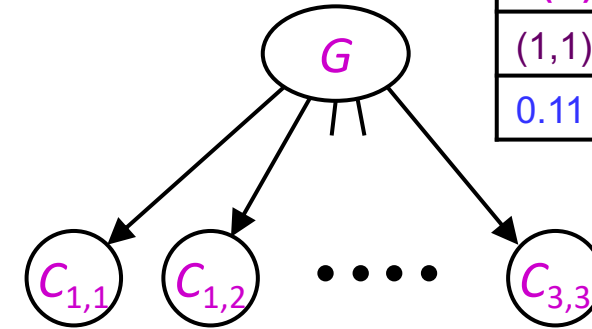
Bayes Net Syntax and Semantics



Bayes Net Syntax



- A set of nodes, one per variable X_i
- A directed, acyclic graph
- A conditional distribution for each node given its **parent variables** in the graph
 - **CPT** (conditional probability table); each row is a distribution for child given values of its parents



$P(G)$			
(1,1)	(1,2)	(1,3)	...
0.11	0.11	0.11	...

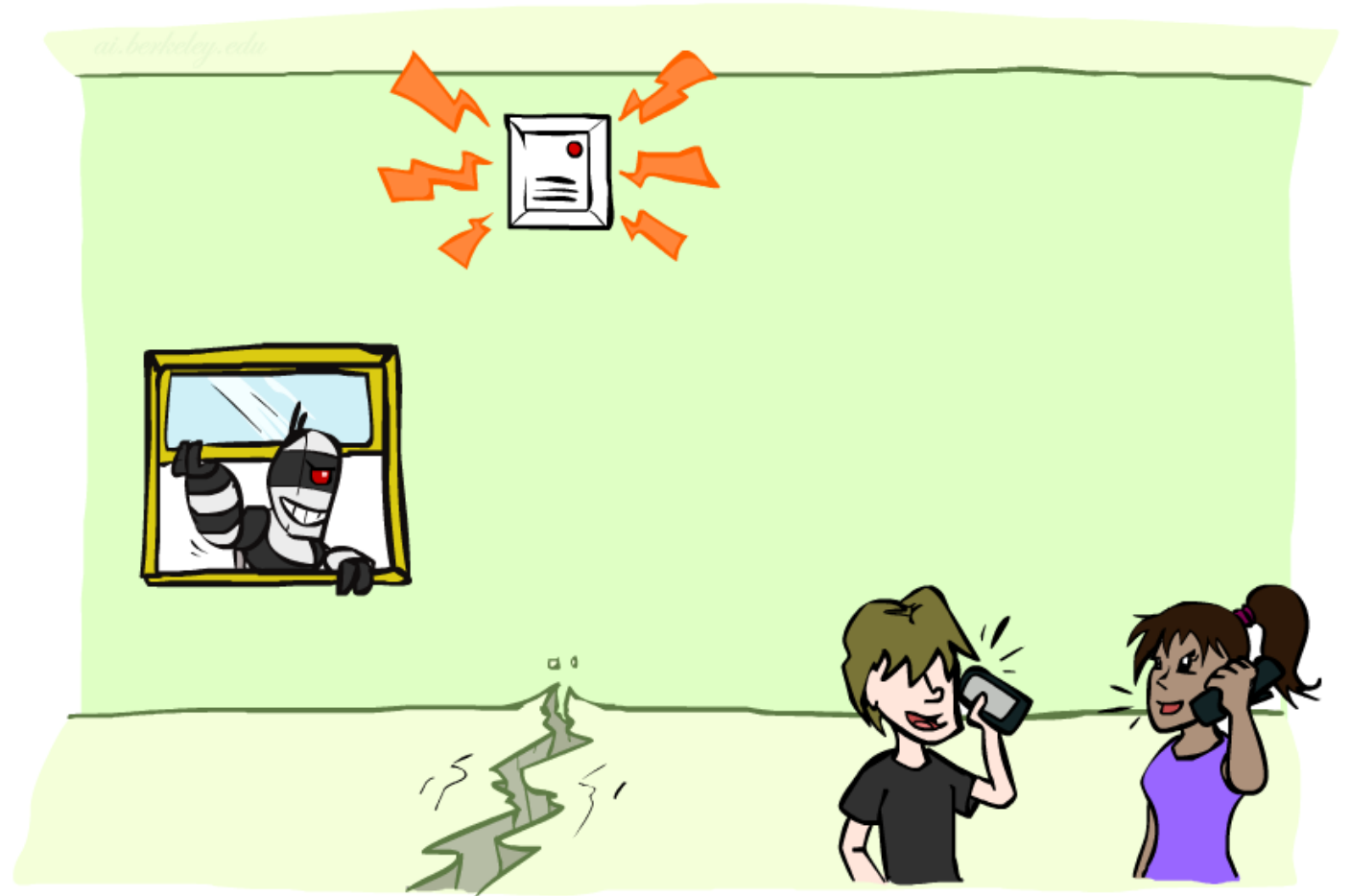
G	$P(C_{1,1} G)$			
	g	y	o	r
(1,1)	0.01	0.1	0.3	0.59
(1,2)	0.1	0.3	0.5	0.1
(1,3)	0.3	0.5	0.19	0.01
...				

Bayes net = Topology (graph) + Local Conditional Probabilities

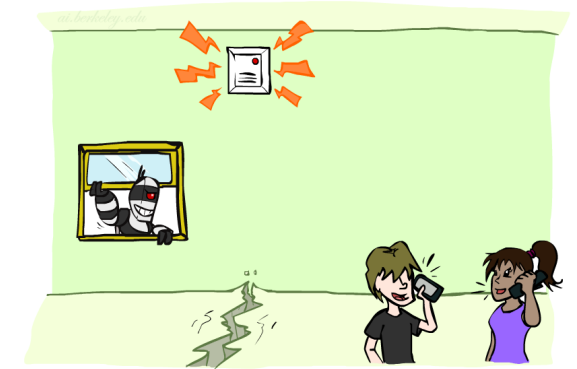
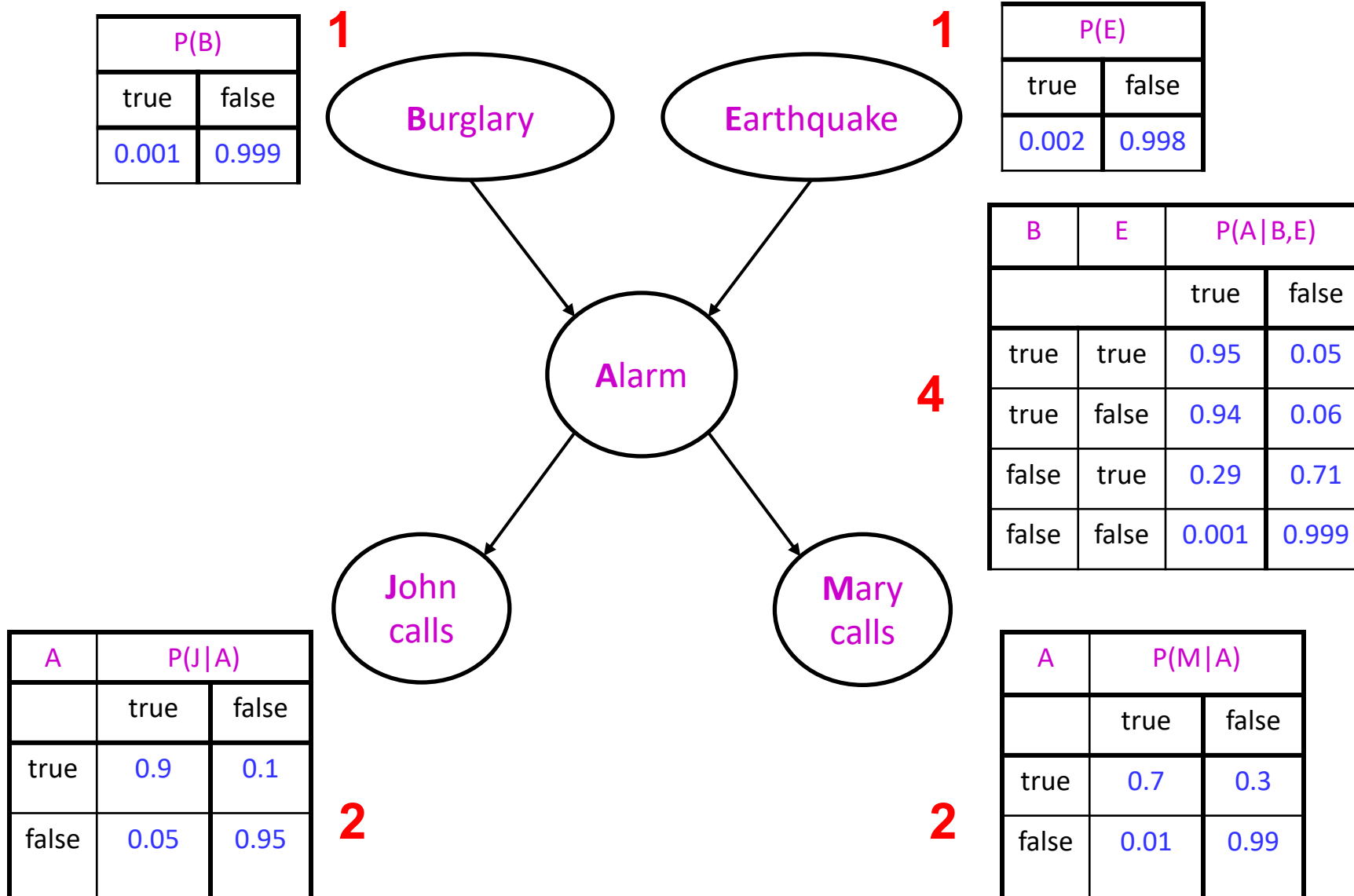
Example: Alarm Network

- Variables

- B: Burglary
- E: Earthquake
- A: Alarm goes off
- J: John calls
- M: Mary calls



Example: Alarm Network



Number of *free parameters* in each CPT:

Parent range sizes $d_1, \dots, d_k$

Child range size d

Each table row must sum to 1

$$(d-1) \prod_i d_i$$

Bayes net global semantics



- Bayes nets encode joint distributions as product of conditional distributions on each variable:

$$P(X_1, \dots, X_n) = \prod_i P(X_i \mid \text{Parents}(X_i))$$

- Exploits sparse structure: number of parents is usually small

Size of a Bayes Net

- How big is a joint distribution over N variables, each with d values?

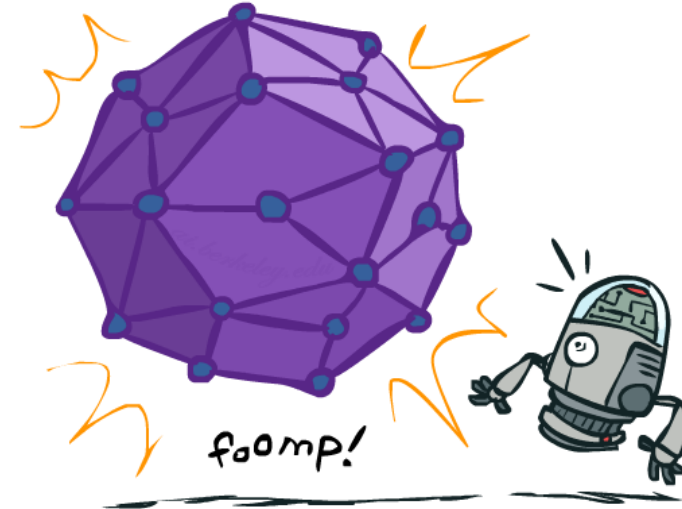
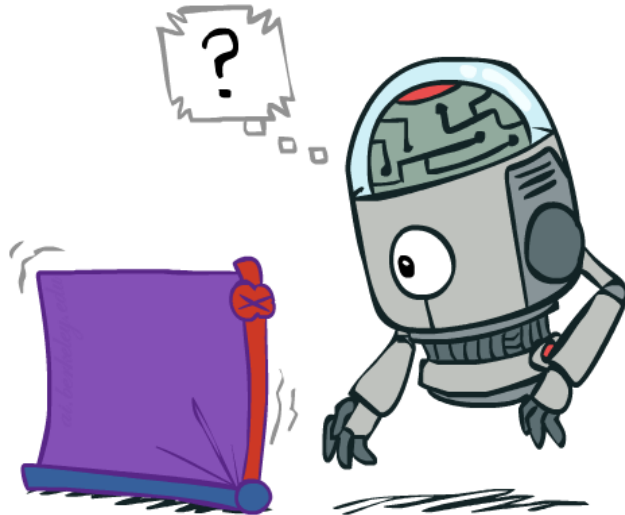
$$d^N$$

- How big is an N -node net if nodes have at most k parents?

$$O(N * d^k)$$

- Both give you the power to calculate $P(X_1, X_2, \dots, X_N)$

- Bayes Nets: huge space savings with sparsity!
- Also easier to elicit local CPTs
- Also faster to answer queries (coming)



Example

P(B)	
true	false
0.001	0.999

P(E)	
true	false
0.002	0.998

$$P(b, \neg e, a, \neg j, \neg m) =$$

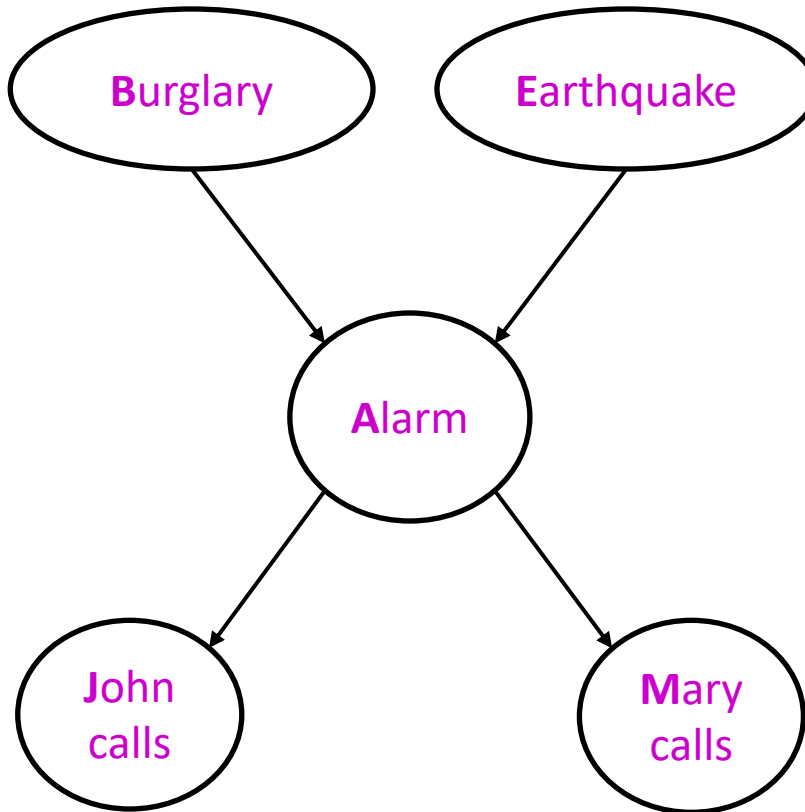
$$P(b) P(\neg e) P(a|b, \neg e) P(\neg j|a) P(\neg m|a)$$

$$= 0.001 \times 0.998 \times 0.94 \times 0.1 \times 0.3 = 0.000028$$

B	E	P(A B,E)	
		true	false
true	true	0.95	0.05
true	false	0.94	0.06
false	true	0.29	0.71
false	false	0.001	0.999

A	P(J A)	
	true	false
true	0.9	0.1
false	0.05	0.95

A	P(M A)	
	true	false
true	0.7	0.3
false	0.01	0.99



Conditional independence in BNs



- Compare the Bayes net global semantics

$$P(X_1, \dots, X_n) = \prod_i P(X_i \mid \text{Parents}(X_i))$$

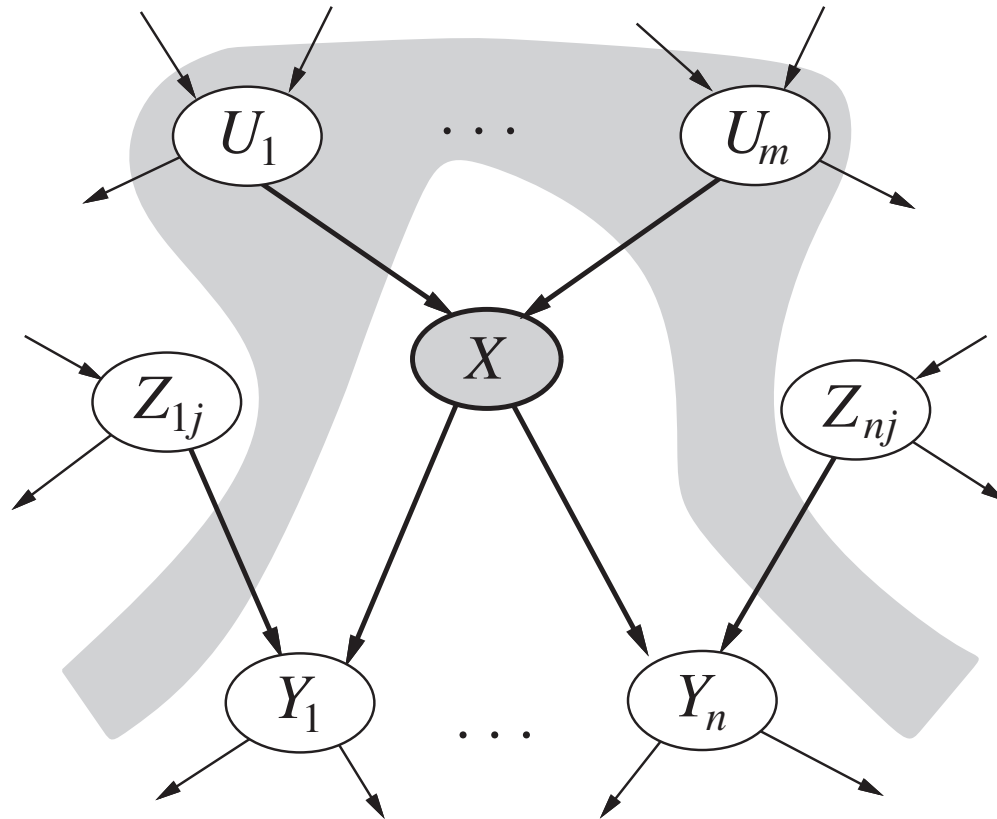
with the chain rule identity

$$P(X_1, \dots, X_n) = \prod_i P(X_i \mid X_1, \dots, X_{i-1})$$

- Assume (without loss of generality) that $X_1, \dots, X_n$ sorted in topological order according to the graph (i.e., parents before children), so $\text{Parents}(X_i) \subseteq X_1, \dots, X_{i-1}$
- So the Bayes net asserts conditional independences $P(X_i \mid X_1, \dots, X_{i-1}) = P(X_i \mid \text{Parents}(X_i))$
 - To ensure these are valid, choose parents for node X_i that “shield” it from other predecessors

Conditional independence semantics

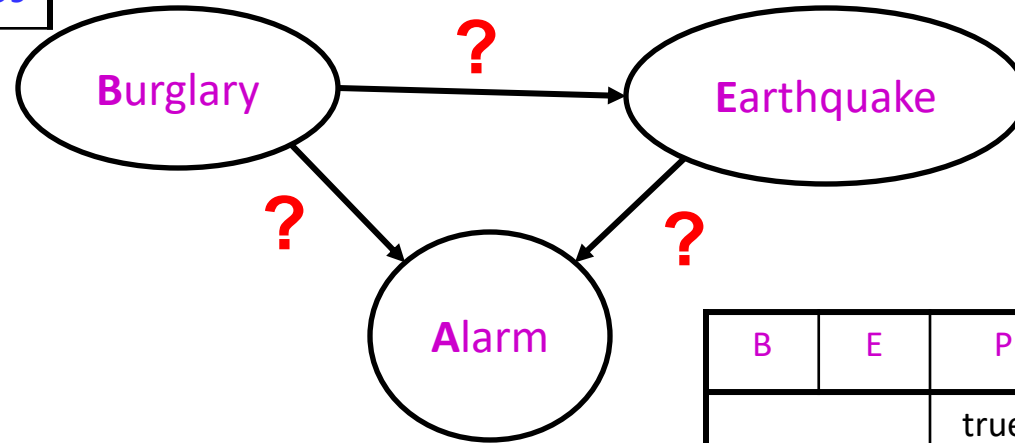
- *Every variable is conditionally independent of its non-descendants given its parents*
- Conditional independence semantics $\Leftrightarrow$ global semantics



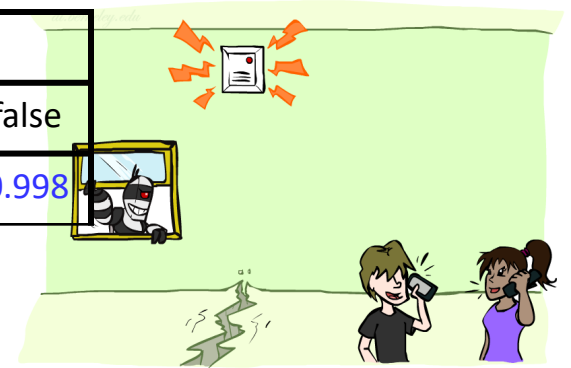
Example: Burglary

- Burglary
- Earthquake
- Alarm

P(B)	
true	false
0.001	0.999



P(E)	
true	false
0.002	0.998

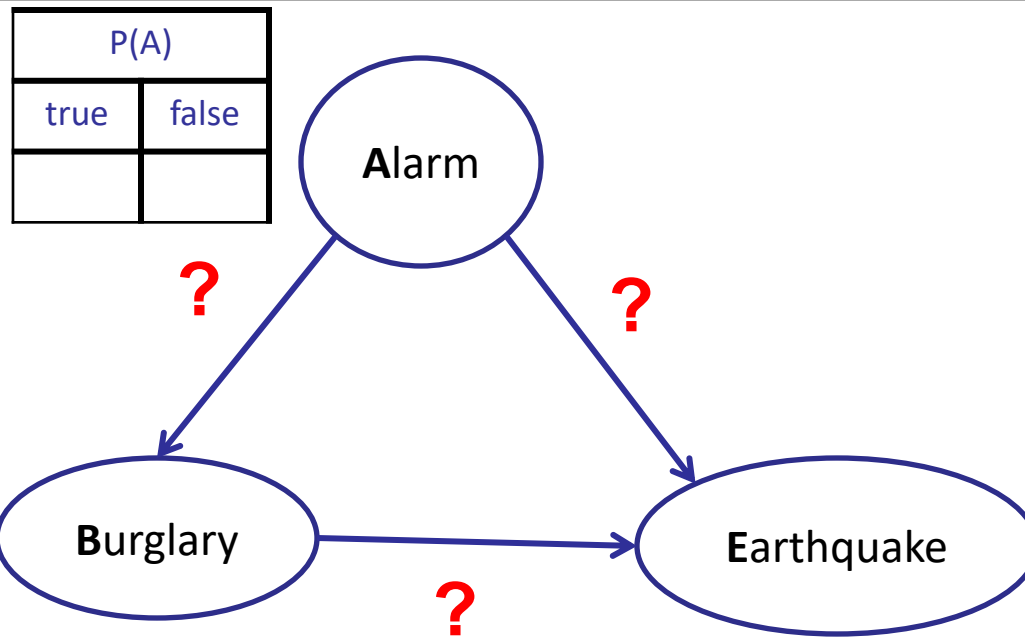


B	E	P(A B,E)	
		true	false
true	true	0.95	0.05
true	false	0.94	0.06
false	true	0.29	0.71
false	false	0.001	0.999

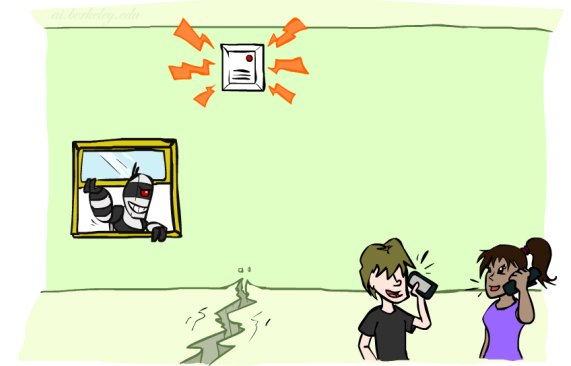
Example: Burglary

- Alarm
- Burglary
- Earthquake

A	P(B A)	
	true	false
true	?	
false		

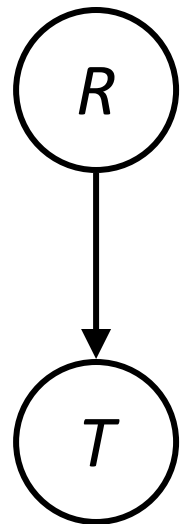
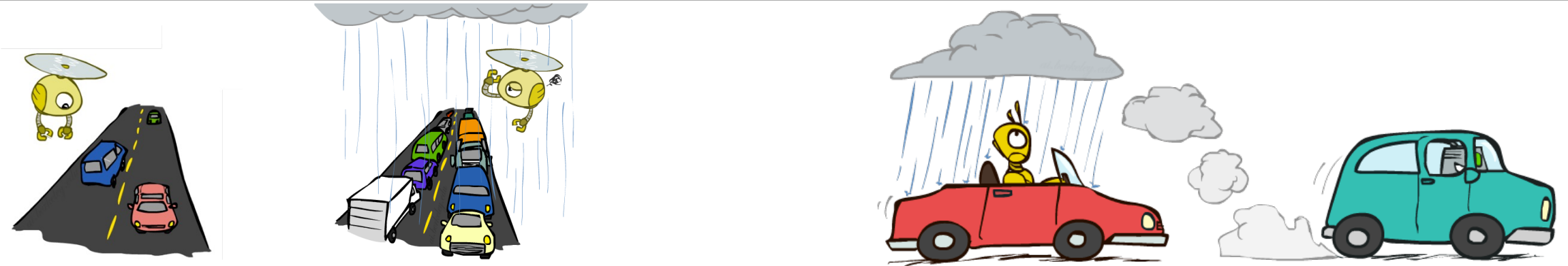


P(A)	
true	false



A	B	P(E A,B)	
		true	false
true	true	?	
true	false		
false	true		
false	false		

Example: Traffic



$P(R)$

+r	1/4
-r	3/4

$P(T|R)$

+r	+t	3/4
	-t	1/4

-r	+t	1/2
	-t	1/2

$P(T, R)$

+r	+t	3/16
+r	-t	1/16
-r	+t	6/16
-r	-t	6/16

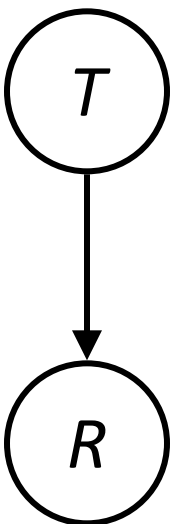
$P(T)$

+t	9/16
-t	7/16

$P(R|T)$

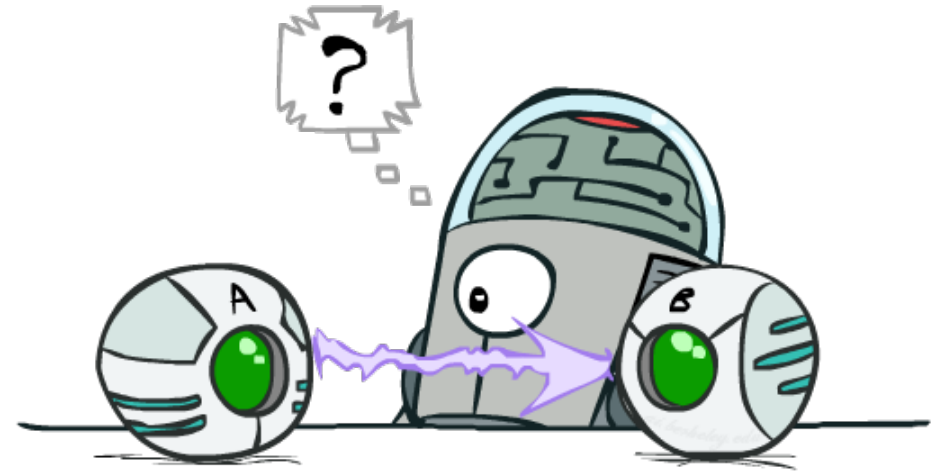
+t	+r	1/3
	-r	2/3

-t	+r	1/7
	-r	6/7



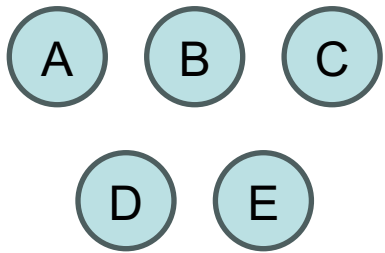
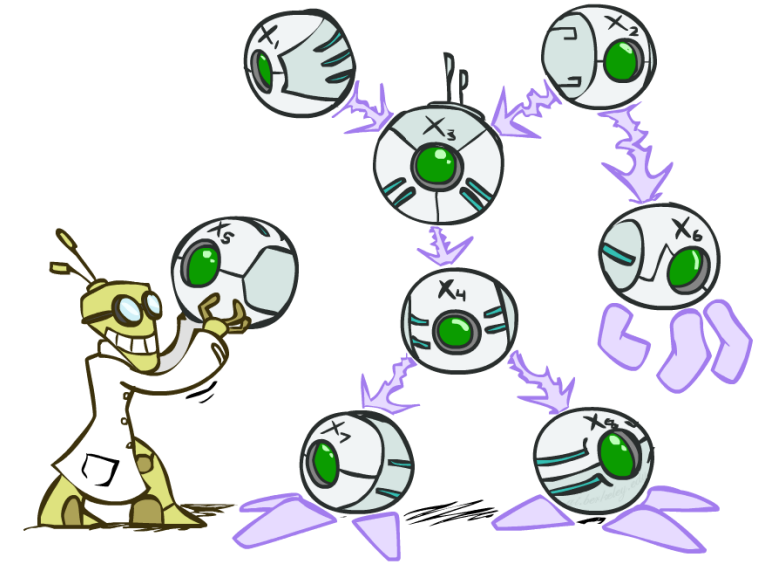
Causality?

- When Bayes' nets reflect the true causal patterns:
 - Often simpler (nodes have fewer parents)
 - Often easier to think about
 - Often easier to elicit from experts
- BNs need not actually be causal
 - Sometimes no causal net exists over the domain (especially if variables are missing)
 - E.g. consider the variables *Traffic* and *Rain*
 - End up with arrows that reflect correlation, not causation
- What do the arrows really mean?
 - Topology may happen to encode causal structure
 - **Topology really encodes conditional independence**
$$P(x_i | x_1, \dots, x_{i-1}) = P(x_i | \text{parents}(X_i))$$

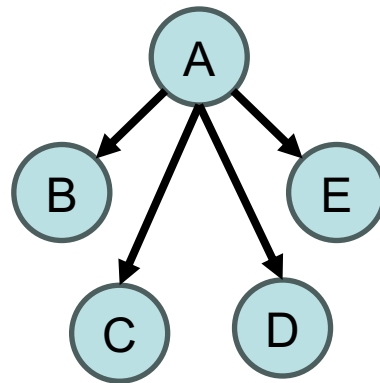


Summary

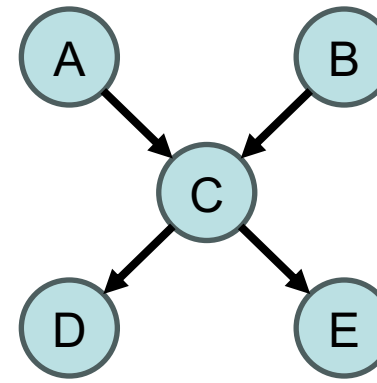
- Independence and conditional independence are important forms of probabilistic knowledge
- Bayes net encode joint distributions efficiently by taking advantage of conditional independence
 - Global joint probability = product of local conditionals
- Allows for flexible tradeoff between model accuracy and memory/compute efficiency



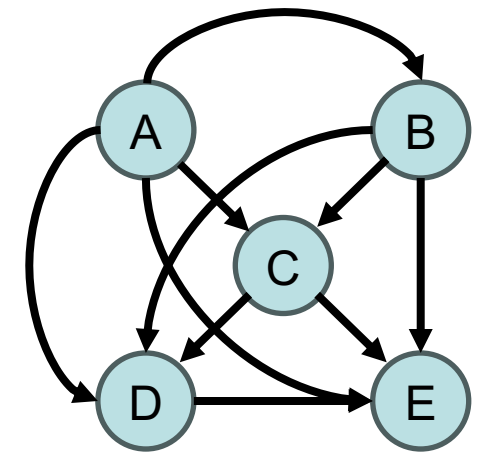
Strict Independence



Naïve Bayes



Sparse Bayes Net



Joint Distribution